

Mission to Mars: Challenge 1

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Team Roles / Timeline

- Leena Dudi:
 - Communications, Data, & Shelter
- Katherine Diaz Govea:
 - Power Generation and Distribution & Thermal Control
- Sergio Long:
 - Human Life Support Consumables & Navigation
- Anthony Nguyen:
 - Propulsion & Colony Size Estimate

Team Strategy

- **Our Challenge:** What minimal set of equipment and its associated mass would be required to set up a sustaining colony of humans on Mars from 2075 to 2125? (launches occurring between 2030-2075)
- **Approach:** Economic rather than Scientific
- **Difficult challenges**
 - Finding reliable sources to make reasonable estimates
 - Evaluating how choosing one system may impact another system
- **How do we tackle them?**
 - Cross-examining with each other to ensure accuracy
 - Predicting improvements in future years by looking at previous trends

Minimum Sustainable Colony Size

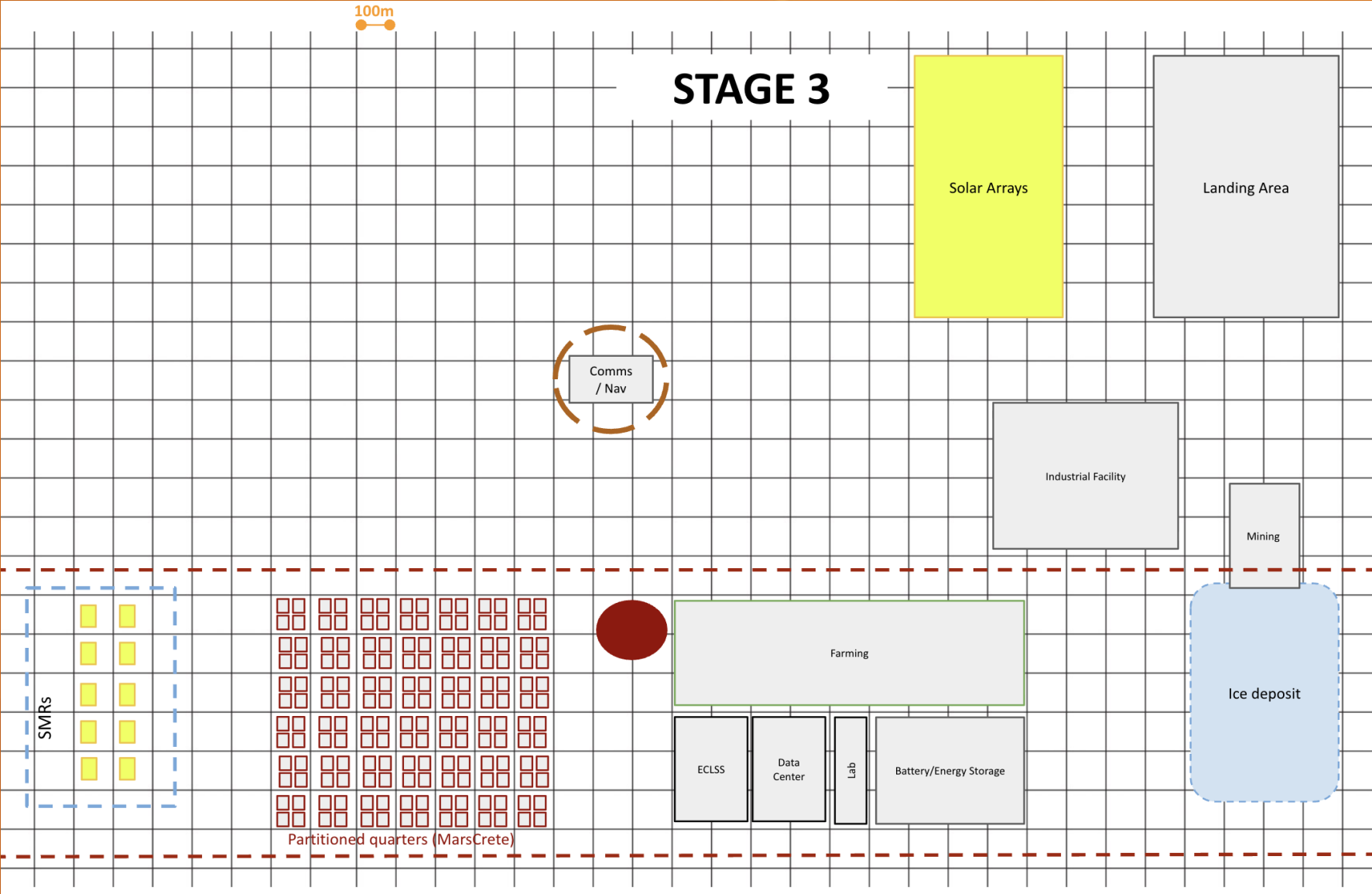
- Key assumption is that intended colony size is roughly 5,000 people
 - Composed of highly skilled workers to maintain infrastructure and provide economic output
 - Population would be allowed to naturally grow over time
- Estimate provided by Chris Impey using studies on Earth, and considers multiple factors (ex. Genetic Diversity & Resilience to Catastrophic events)
- Some studies estimate smaller numbers for sustainability, but we wanted to aim for a city-level colony that would last in the long-term given that we can use advanced reusable launch vehicles

Getting to Mars (2030 - 2075)

- Rocket: New Glenn 9x4 (*with orbital refueling) for 70t payload capacity to Mars

Phase	Windows	Total Flights	Flights / Window	People / Flight	Population Added	Total Phase Payload	Mass / Person
Stage 1	1–4	28	7	~7	200	2,050 t	10.25 t
Stage 2	5–12	49	6–7	~33	1600	3,410 t	2.13 t
Stage 3	13–20	68	8–9	~47	3200	2,000 t	0.63 t
TOTALS	20	145	~7.25 (avg)	~34 (avg)	5,000	7,460 t	1.49 t (avg)

Settlement Map



Propulsion

- In total we need 145 launches for our approach
- DRM estimates each total mission mass to be 800 – 1200 metric tons, most of it being for propulsion (we assume 80%)
- Taking the middle value, we need 116,000 metric tons for propellant and propellant hardware
- Since time to Mars is already around 200 days, we decided against electric propulsion due to low thrust
- Sticking to chemical propulsion for reliability in reaching required delta V of 12-14 km/s and reasonable time of flight

Living on Mars

- **Shelter**

- Challenges: internal air pressure, space radiation, extreme temperatures
- Solution:
 - 2030 – 2060: Sierra Space LIFE 1400 module (fits 4-6 people)
 - 2060+: Lava Tubes with MarsCrete reinforcement
- Potential locations: Hebrus Valles (~16 skylights), Arsia Mons (~7 skylights)

- **Thermal Control**

- Challenges: Extreme temperatures, thin atmosphere, dust storms
- Solution (prevents overheating/freezing, reuses waste heat):
 - Passive Systems: radiators, multi-layer insulation, regolith burial
 - Active Systems: electric heaters, pumped fluid loops

Infrastructure/Habitation	
Unit	Mass
LIFE1400 hubs	1200 t
Atmospheric Sealing	350 t
Thermal & Fluid Distribution	90 t
Lava Tube Operations	200 t
Cave Partioning	210 t
Total = 2050 t	

Industrial & Manufacturing

- **Resource Extraction:** mining of ice, hematite (iron), & high-purity silica (glass)
- **MarsCrete Reinforcement:** sealing of Lava Tubes using extracted basalt/hematite
- **Propellant Plant:** Sabatier reactors that synthesize methane/oxygen for transportation fuel
- **Export:** Extract & refine Deuterium, metals, & opals for trade
- **Closed-Loop Sustainability:** machine shop for repair; chemical synthesis to upcycle waste into sealants & binders

Industrial	
Unit	Mass
Mining & Export Extraction	450 t
Metals & Glass Refinement	600 t
Chemical & Propellant Plant	450 t
MarsCrete Reinforcement Equipment	250 t
Machine Shop/Maintenance	100 t
Total = 1850 t	

Human Life Support Consumables

- Closed-loop systems for self-reliance on air, water, & waste management (i.e. ISS Environmental Control and Life Support Systems)
- Water - Reclaim 98% of wastewater using current ISS distillation
- Oxygen – Similar system will be used for ECLSS
 - ISS Oxygen Generation Assembly (electrolyze water into O₂ and H₂)
 - Carbon Dioxide Reduction Assembly (react H₂ with CO₂ into methane and water)
- Food – Freeze-dried foods for space mission to Mars, on-site food production necessary for colony growth

Life Support	
Unit	Mass
Atmospheric Management	250 t
Water/Waste Loop	200 t
Beginning Food Supply	350 t
Farming Equipment	350 t
Health Equipment	150 t
Total = 1300 t	

Power Generation and Distribution

- We need a continuous power system to support our colony
- Total Power Demand estimated to be 42 MW

Primary: 5 MW small modular nuclear reactors

- Can provide power 24/7, unaffected by dust storms

Secondary: large scale solar-arrays

- Cannot be used at night or during dust storm but provides extra power as the colony grows

Power Demand	
Industrial / ISRU	20 MW
Agriculture	8 MW
Thermal Control	3 MW
Life Support	5 MW
Lighting & Internal	3 MW
Data, Comms, Nav	3 MW
TOTAL DEMAND	42 MW

Power	
Unit	Mass
SMRs	550 t
Solar Arrays	80 t
Power Distribution Hubs	40 t
Thermal Radiators	120 t
Cabling/Wiring	35 t
Switchgear & Transformers	25 t
Total = 850 t	

Data, Communications, & Navigation

- **Communications: 3 layered system**
 - UHF (orbital), S-band (direct to Earth), Ka-Band (high-speed video)
 - Ka-band limited to 3.5 hours/day per person (reduces peak power demand and only requires 4 High-Throughput satellites)
- **Data:** To bypass 20-min communication delay, data center that stores a very basic information hub will be established
- **Navigation:**
 - Autonomous systems needed for position-tracking due to communication delay: celestial tracking & relay satellites
 - PNT technology & pseudolites will be implemented due to lack of GPS on Martian surface
 - Terrain Relative Navigation will be used to improve precision of landing

Comms, Data, & Nav	
Unit	Mass
HTS Satellite Constellation	60 t
Data Center	120 t
Ground Station	35 t
Local Nav (Pseudolites)	15 t
Internal Fiber/Li-Fi	30 t
Total = 260 t	

Total Mass

System	Mass
Infrastructure & Habitation	2,050 t
Industrial & Manufacturing	1,850 t
Life Support (ECLSS)	1,300 t
Energy & Power Generation	850 t
Comms, Data, & Navigation	260 t
Settlement Payload = 6,310 t	
Transit Payload	1150 t
GRAND TOTAL Payload = 7,460 t	

